7. A picture is painted on a ceiling of a marriage hall whose length and breadth are 18 *m* and 7 *m* respectively. There is a border of 10 *cm* along each of its sides. Find the area of the border.

8. A canal of width 1 *m* is constructed all along inside the field which is 24 *m* long and 15 *m* wide. Find (i) the area of the canal (ii) the cost of constructing the canal at the rate of ₹12 per *sq.m*.

Objective type questions

9. The formula to find the area of the circular path is

   (i) $\pi(R^2 - r^2)$ *sq. units* (ii) πr^2 *sq. units* (iii) $2\pi r^2$ *sq. units* (iv) $\pi r^2 + 2r$ *sq. units*

10. The formula used to find the area of the rectangular path is

    (i) $\pi(R^2 - r^2)$ *sq. units* (ii) $(L \times B) - (l \times b)$ *sq. units* (iii) *LB sq. units* (iv) *lb sq. units*

11. The formula to find the width of the circular path is

    (i) $(L - l)$ *units* (ii) $(B - b)$ *units* (iii) $(R - r)$ *units* (iv) $(r - R)$ *units*

Exercise 2.4

Miscellaneous Practice problems



1. A wheel of a car covers a distance of 3520 *cm* in 20 rotations. Find the radius of the wheel?

2. The cost of fencing a circular race course at the rate of ₹8 per metre is ₹2112. Find the diameter of the race course.

3. A path 2 *m* long and 1 *m* broad is constructed around a rectangular ground of dimensions 120 *m* and 90 *m* respectively. Find the area of the path.

4. The cost of decorating the circumference of a circular lawn of a house at the rate of ₹55 per metre is ₹16940. What is the radius of the lawn?

5. Four circles are drawn side by side in a line and enclosed by a rectangle as shown below.

   If the radius of each of the circles is 3 *cm*, then calculate:

   (i) The area of the rectangle.
   (ii) The area of each circle.
   (iii) The shaded area inside the rectangle.



Challenge Problems

6. A circular path has to be constructed around a circular lawn. If the outer and inner circumferences of the path are 88 *cm* and 44 *cm* respectively, find the width and area of the path.

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